

## TRI-COUNTY REGIONAL, WI, USA

WMO: 726416

Lat: 43.212N

Lon: 90.181W

Elev: 219

StdP: 98.73

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 14921

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-23.8 | (c)<br>-20.5 | (d)<br>-27.6                  | (e)<br>0.3 | (f)<br>-22.6 | (g)<br>-25.0 | (h)<br>0.4 | (i)<br>-19.9 | (j)<br>10.6           | (k)<br>-5.9 | (l)<br>9.1 | (m)<br>-6.3 | (n)<br>1.5            | (o)<br>260 | (p)<br>0.524 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |            |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD       |
| (a)<br>7      | (b)<br>11.7               | (c)<br>32.3     | (d)<br>23.6 | (e)<br>30.9 | (f)<br>22.8 | (g)<br>29.2 | (h)<br>21.8 | (i)<br>25.2         | (j)<br>30.3 | (k)<br>24.1 | (l)<br>28.9 | (m)<br>23.2 | (n)<br>27.6 | (o)<br>4.3              | (p)<br>190 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.6                            | 18.9 | 28.3 | 22.6 | 17.8 | 27.1 | 21.7 | 16.9 | 26.2 | 78.0          | 30.2 | 73.9 | 29.1 | 69.9 | 27.6 | 28.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 9.4           | (b) 8.4 | (c) 7.5 | (e) -29.1                  | (f) 34.6 | (g) 3.7            | (h) 1.7 | (i) -31.8                                          | (j) 35.9 | (k) -33.9  | (l) 36.9 | (m) -36.0  | (n) 37.9 | (o) -38.6  | (p) 39.1 |    |    |
| (a) 9.4           | (b) 8.4 | (c) 7.5 | (e) -29.0                  | (f) 26.9 | (g) 3.1            | (h) 1.0 | (i) -31.3                                          | (j) 27.7 | (k) -33.1  | (l) 28.3 | (m) -34.8  | (n) 28.8 | (o) -37.1  | (p) 29.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.6    | -7.1  | -5.7  | 1.7   | 8.7   | 15.0  | 20.4  | 22.4  | 21.2  | 17.2  | 9.9   | 3.0   | -4.2  |  |  |
|                                                                        | DBStd       | 11.66  | 6.94  | 6.75  | 6.55  | 5.28  | 4.81  | 3.51  | 3.07  | 3.04  | 4.39  | 5.08  | 5.41  | 6.67  |  |  |
|                                                                        | HDD10.0     | 2061   | 531   | 440   | 269   | 84    | 10    | 0     | 0     | 0     | 3     | 65    | 218   | 441   |  |  |
|                                                                        | HDD18.3     | 3924   | 789   | 673   | 518   | 290   | 127   | 19    | 4     | 9     | 72    | 266   | 460   | 699   |  |  |
|                                                                        | CDD10.0     | 1553   | 0     | 0     | 11    | 46    | 165   | 311   | 383   | 346   | 219   | 63    | 8     | 1     |  |  |
|                                                                        | CDD18.3     | 375    | 0     | 0     | 1     | 2     | 24    | 80    | 129   | 96    | 38    | 4     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 3627   | 0     | 0     | 8     | 42    | 269   | 774   | 1266  | 844   | 375   | 49    | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 1141   | 0     | 0     | 1     | 8     | 78    | 244   | 446   | 249   | 107   | 8     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.0    | 3.3   | 3.4   | 3.5   | 4.0   | 3.4   | 2.7   | 2.2   | 2.1   | 2.5   | 3.1   | 3.2   | 3.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 898    | 24    | 30    | 52    | 98    | 105   | 145   | 107   | 115   | 85    | 57    | 54    | 42    |  |  |
|                                                                        | PrecMax     | 1227   | 50    | 63    | 108   | 160   | 275   | 277   | 227   | 462   | 197   | 125   | 160   | 95    |  |  |
|                                                                        | PrecMin     | 610    | 7     | 9     | 14    | 38    | 39    | 32    | 43    | 28    | 14    | 19    | 7     | 6     |  |  |
|                                                                        | PrecStd     | 161    | 13    | 16    | 30    | 37    | 52    | 63    | 51    | 99    | 45    | 30    | 38    | 23    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 9.4   | 13.8  | 24.3  | 27.7  | 32.3  | 33.5  | 35.5  | 33.0  | 32.0  | 27.7  | 20.9  | 14.0  |  |  |
|                                                                        |             | MCWB   | 5.7   | 9.8   | 16.9  | 17.1  | 21.9  | 23.7  | 24.7  | 24.1  | 22.8  | 19.2  | 14.3  | 12.4  |  |  |
|                                                                        | 2%          | DB     | 6.2   | 8.0   | 18.1  | 23.9  | 28.8  | 31.3  | 32.4  | 31.2  | 29.7  | 24.2  | 17.1  | 8.9   |  |  |
|                                                                        |             | MCWB   | 3.3   | 5.1   | 12.6  | 15.4  | 19.3  | 22.6  | 24.2  | 23.4  | 21.9  | 17.9  | 12.1  | 6.8   |  |  |
|                                                                        | 5%          | DB     | 3.6   | 5.3   | 14.9  | 21.3  | 26.4  | 29.4  | 30.8  | 29.3  | 27.5  | 21.7  | 14.2  | 6.3   |  |  |
|                                                                        |             | MCWB   | 2.0   | 2.8   | 10.3  | 13.6  | 18.2  | 21.4  | 23.2  | 22.2  | 20.2  | 16.4  | 10.0  | 4.4   |  |  |
|                                                                        | 10%         | DB     | 2.1   | 2.9   | 11.1  | 18.6  | 23.9  | 27.7  | 29.0  | 27.7  | 25.4  | 18.8  | 11.6  | 4.0   |  |  |
|                                                                        |             | MCWB   | 0.8   | 0.8   | 7.0   | 11.7  | 17.0  | 20.7  | 22.3  | 21.3  | 19.0  | 14.1  | 8.3   | 2.2   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 6.1   | 9.2   | 17.4  | 18.5  | 22.9  | 25.6  | 26.8  | 25.9  | 24.5  | 21.8  | 15.1  | 12.8  |  |  |
|                                                                        |             | MCDB   | 8.2   | 13.1  | 24.0  | 24.5  | 30.3  | 30.9  | 32.0  | 31.0  | 29.8  | 25.3  | 19.1  | 14.3  |  |  |
|                                                                        | 2%          | WB     | 3.7   | 5.5   | 14.4  | 16.3  | 21.1  | 24.2  | 25.5  | 24.7  | 22.9  | 19.6  | 13.1  | 6.9   |  |  |
|                                                                        |             | MCDB   | 5.6   | 7.6   | 17.5  | 22.5  | 26.5  | 29.2  | 30.7  | 29.1  | 27.4  | 22.4  | 16.4  | 8.2   |  |  |
|                                                                        | 5%          | WB     | 2.1   | 3.0   | 10.3  | 14.7  | 19.7  | 22.9  | 24.4  | 23.6  | 21.6  | 17.3  | 10.7  | 4.5   |  |  |
|                                                                        |             | MCDB   | 3.7   | 5.1   | 14.2  | 20.2  | 24.5  | 28.1  | 29.4  | 27.6  | 25.8  | 20.7  | 13.7  | 6.2   |  |  |
|                                                                        | 10%         | WB     | 0.7   | 1.1   | 7.4   | 12.4  | 18.1  | 21.6  | 23.5  | 22.5  | 20.4  | 14.6  | 8.3   | 2.5   |  |  |
|                                                                        |             | MCDB   | 1.9   | 2.6   | 10.9  | 17.8  | 22.4  | 26.1  | 27.7  | 26.4  | 23.9  | 18.2  | 11.0  | 4.0   |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 9.5   | 10.0  | 10.4  | 12.0  | 12.0  | 11.5  | 11.7  | 11.5  | 12.5  | 11.3  | 9.7   | 8.6   |  |  |
|                                                                        | 5% DB       | MCDBR  | 10.4  | 11.6  | 14.2  | 16.4  | 15.0  | 13.8  | 13.3  | 13.1  | 14.7  | 15.0  | 14.0  | 10.1  |  |  |
|                                                                        |             | MCWBR  | 8.3   | 8.9   | 9.0   | 9.3   | 7.2   | 6.6   | 6.2   | 6.3   | 7.5   | 8.9   | 9.8   | 8.4   |  |  |
|                                                                        | 5% WB       | MCDBR  | 9.6   | 11.4  | 13.0  | 14.0  | 12.3  | 11.6  | 11.2  | 10.8  | 11.9  | 12.5  | 12.7  | 9.5   |  |  |
|                                                                        |             | MCWBR  | 7.9   | 9.0   | 9.0   | 8.8   | 6.8   | 6.3   | 6.2   | 6.0   | 6.7   | 8.9   | 9.8   | 8.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.274 | 0.308 | 0.326 | 0.372 | 0.399 | 0.419 | 0.422 | 0.414 | 0.376 | 0.336 | 0.296 | 0.275 |  |  |
|                                                                        | taud        |        | 2.383 | 2.290 | 2.391 | 2.306 | 2.280 | 2.261 | 2.257 | 2.278 | 2.367 | 2.476 | 2.512 | 2.438 |  |  |
|                                                                        | Ebn at Noon |        | 867   | 884   | 916   | 893   | 874   | 854   | 847   | 842   | 850   | 848   | 839   | 834   |  |  |
|                                                                        | Edh at Noon |        | 77    | 101   | 105   | 124   | 131   | 134   | 133   | 126   | 107   | 84    | 69    | 67    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.75  | 2.81  | 3.83  | 4.54  | 5.39  | 5.99  | 6.08  | 5.26  | 4.30  | 2.77  | 1.82  | 1.34  |  |  |
|                                                                        | RadStd      |        | 0.21  | 0.27  | 0.45  | 0.39  | 0.40  | 0.45  | 0.33  | 0.35  | 0.35  | 0.34  | 0.15  | 0.12  |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (52 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page